

SIDDARTH DAYASAGAR

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EDUCATION

Northeastern University, Boston, MA

Expected Dec 2026

Master's in Robotics (Electrical and Computer Engineering)

CGPA 3.63

Relevant Coursework:

Mobile Robotics, Robotics Mechanics and Control, Control Systems Engineering, Formal methods for dynamic systems, Legged Robotics, Pattern Recognition and Computer Vision

Jain University, Bangalore, India

Jun 2023

Bachelor of Technology in Robotics and Automation

CGPA 8.66

SKILLS

- **Programming:** Python, C++, MATLAB
- **Libraries:** ROS (noetic), OpenCV, PCL, ROS2 (humble, Jazzy), CasADi, Pinocchio
- **Design and Development Tools:** Fusion 360, Ultimaker CURA, Arduino UNO R3 and Raspberry Pi 4B+
- **Simulation:** Gazebo Classic, Gazebo harmonic, Mujoco, IsaacSim, IsaacLab
- **Certifications:** Udacity Robotics Nanodegree, Rigbetel Labs ROS Mentorship Program

WORK EXPERIENCE

Robotics Research Intern

Aug 2025 – Dec 2025

SpaceData Inc., Tokyo, Japan

- **Engineered** a PPO-based quadruped locomotion policy in Isaac Sim, improving performance by approximately **70% over NVIDIA's pretrained baseline** by refining rewards, optimising hyperparameters, and implementing domain randomisation.
- **Integrated** ROS2 Nav2 with an **MPPI controller** to establish a robust indoor autonomy pipeline, enabling global planning, obstacle-aware local control, and multi-sensor navigation for disaster-response scenarios.
- **Pioneered** a next-generation outdoor navigation system by building a **grid map**-driven autonomy pipeline and designing an **MPC controller** that extends quadruped navigation from controlled indoor labs to large-scale, unstructured outdoor environments, enabling multi-terrain deployment for disaster-response missions.
- **Architected Space Station OS**, a modular ROS 2 autonomy framework that combined subsystems into a unified control architecture, using behaviour trees, system diagnostics, and multi-node coordination to create a scalable, fault-tolerant robotics simulation platform.

Robotics Intern

Sept 2022 – Apr 2023

Flomobility, Bangalore, India

- **Architected and deployed** a vision-based localisation pipeline for autonomous lawn-mower robots using **AprilTag boards**, achieving **high-precision pose estimation with negligible localisation error**.
- **Implemented and tuned** Bang-Bang and PID control strategies for actuator control, **enhancing system stability, responsiveness, and overall operational efficiency**.
- **Collaborated** cross-functionally with mechanical, electrical, and software teams to validate end-to-end autonomy during on-robot testing and deployment.

PROJECTS

Learning-Based Extension of Classical Pick-and-Place using PPO

Nov 2025 - Dec 2025

- **Extended** a classical pick-and-place manipulation pipeline by training a **PPO** policy for a **7-DoF Franka Panda arm** in MuJoCo, using Cartesian end-effector action increments instead of explicit inverse kinematics.
- **Implemented** training pipeline in **PyTorch** with Gaussian policies, GAE, clipped objectives, and entropy regularisation for stable learning in continuous action spaces.
- **Designed** task-specific reward shaping and safety penalties (reach-to-object, joint limits, smooth motion, collision avoidance) and validated pick-and-place performance using MuJoCo.

Autonomous Racing F1/10

May 2025- Sept 2025

- **Architected** a full-stack trajectory tracking and optimisation framework on the F1TENTH platform using **both Model Predictive Control (MPC) and Model Predictive Contouring Control (MPCC)** in ROS2 and CasADi
- Formulated a nonlinear kinematic bicycle model and embedded real-time constraints including velocity, steering angle, and curvature bounds; achieved stable control up to **4.5 m/s** on hairpin curves with less than **5% cross-track error**
- **Designed safety-aware cost functions and hard constraints** (track boundaries, obstacle clearance, actuator limits) within the optimisation loop, improving robustness and reducing off-track violations during lap-level evaluations.

MPC-Based Missile GNC

Mar 2025- Apr 2025

- **Engineered** a high-fidelity missile guidance, navigation, and control (GNC) system using **Model Predictive Control (MPC)** and **Linear Quadratic Regulation (LQR)** in MATLAB with CasADi
- **Benchmarked** MPC's performance against LQR in constrained environments, showcasing **MPC's superior ability to handle state/input limits**, mitigate disturbances, and optimise manoeuvrability
- **Implemented** real-time constraint handling and predictive horizon tuning, **improving pitch stability by 45%** and **reducing control effort by 38%** compared to the baseline